CFRP Laminate Plies, Layups and ABD Matrices

Claude Code chat transcript — ME295 project, Code for Project/10-23-26 (Tara)

User

I have a CFRP material with E1 = 140 GPa, E2 = 10 GPa, G12 = 6 GPa, and nu12 = 0.34. Use the agent PlyPropertyGenerator to create material properties in OpenSeesPy for plies at 0 and 90 degrees to the global xy coordinate sytsem. Then use the agent LayupFromPlyGenerator to write the OpenSeesPy code for a layup with a 0-degree ply on top and a 90-degree ply on the bottom, where both plies have a 1 mm thickness

Claude

The PlyPropertyGenerator agent is now building the 0° and 90° ply materials. Once it finishes, I'll pass its output to LayupFromPlyGenerator to build the two-ply layup: the 90° ply on the bottom and the 0° ply on top, each 1 mm thick.

Claude

The ply materials are done, and the LayupFromPlyGenerator agent is now building the [90/0] laminate. I'll report back when it finishes.

So far, the 0° and 90° materials are in `ply_materials_0_90.py`. The script runs cleanly in OpenSeesPy 3.8. A check on a single element matched the expected stiffness to within rounding error for both angles.

- **Units:** N, mm, MPa, so your inputs were converted to E1 = 140000, E2 = 10000 and G12 = 6000 MPa.
- **v23:** you didn't give it, so it's assumed equal to v12 = 0.34. That gives G23 ≈ 3731 MPa.
- **Material tags:**
  - 0° uses tags 1/2/3 (ElasticOrthotropic, then PlaneStress, then PlateFromPlaneStress).
  - 90° uses tags 4/5/6.
- **Applying the ply angle:** the two material sets are identical. In OpenSeesPy 3.8 the angle is applied only through the element's `-local` axis: `1,0,0` for 0° and `0,1,0` for 90°.
- **Mixing angles in one layup:** a single shell element can have only one orientation, so its layers can't have different angles. The 0/90 laminate therefore needs one element per ply, stacked at the same nodes, which is the approach in the existing `layup.py`. I've told the layup agent to use that.

Claude

The ply materials and the two-ply layup code are both written and work. I re-ran the layup script myself, and its laminate stiffness matches classical lamination theory.

What was written:

- `ply_materials_0_90.py` has the 0° and 90° materials.
  - 0°: material tags 1/2/3, element orientation `'-local', 1, 0, 0`.
  - 90°: material tags 4/5/6, element orientation `'-local', 0, 1, 0`.
  - The two material sets are identical. OpenSeesPy has no material that takes a ply angle, so the angle comes only from the element's `-local` direction.
- `layup_90_0.py` has the layup: 90° ply on the bottom (z = −1 to 0 mm) and 0° ply on top (z = 0 to 1 mm).
  - One element can only have one orientation, so each mesh face gets two elements on the same four nodes, one per ply.
  - Each ply's section is 2 mm thick, with empty layers filling the part where the other ply sits. This puts each ply at the right height.
  - It reuses the existing `layup.py` and `ply_properties.py` rather than copying them.

Stiffness check (units N/mm, N, N·mm):

	Classical lamination theory	OpenSeesPy	Max error
A11 = A22	151249	151249	~1e-16
B11 / B22	+65541 / −65541	same	~1e-16
D11 = D22	50416	50219	0.39%

- **Bending (D) is slightly low.** That's expected with 8 layers per ply. Running with `--n-sub 32` brings the error down to 0.02%.
- **Bending–stretching coupling (B) is not zero.** That's correct for an unsymmetric layup like this one: in-plane loads will also bend the plate.

Assumptions and limits:

- **Units and v23:** inputs were converted to MPa (E1 = 140000, E2 = 10000, G12 = 6000). v23 wasn't given, so it's assumed equal to v12 = 0.34. It doesn't affect A, B or D.
- **Elements must use `-noeas`.** Without that option the two stacked elements don't add up correctly. The cost is that elements are stiffer on coarse meshes, so refine the mesh until results stop changing.
- **Two elements per face** doubles the element count.
- **Material tags 1–7 are taken,** so other materials in your model need tags of 8 or higher.
- **Not built yet:** no mesh, boundary conditions or analysis.

User

Not now

Claude

Okay, I've left everything uncommitted. The new scripts `ply_materials_0_90.py` and `layup_90_0.py` are in the `10-23-26 (Tara)` folder whenever you want to commit them.

User

Now use the agent ABDFromOpenSeesPyLayup to calculate the full ABD matrix from the OpenSeesPy code and report it to me in this chat

Claude

The ABDFromOpenSeesPyLayup agent is now working out the full ABD matrix from the OpenSeesPy code in `layup_90_0.py`. I'll post the A, B and D matrices here when it finishes.

Claude

Here is the full ABD matrix of the [90/0] layup, measured directly from the OpenSeesPy code in `layup_90_0.py`. It matches the ABD the layup script printed earlier.

The relations are N = A ε + B κ and M = B ε + D κ. The strains are [ε_xx, ε_yy, γ_xy], with engineering shear strain, and the curvatures are [κ_xx, κ_yy, κ_xy].

A [N/mm]

	ε_xx	ε_yy	γ_xy
N_xx	151248.88	6856.62	0
N_yy	6856.62	151248.88	0
N_xy	0	0	12000

B [N]

	κ_xx	κ_yy	κ_xy
N_xx	65541.18	0	0
N_yy	0	-65541.18	0
N_xy	0	0	0

D [N·mm]

	κ_xx	κ_yy	κ_xy

M_xx	50219.36	2276.61	0
M_yy	2276.61	50219.36	0
M_xy	0	0	3984.375

All the terms shown as zero are effectively zero (below 1e-11). The matrix is symmetric to within rounding error.

How it was measured: the agent put the two stacked shell elements (one per ply) on the same four nodes of a 1 × 1 mm square. It then applied each of the six strains and curvatures one at a time and read off the resulting forces and moments. No lamination-theory formulas were used.

What the numbers mean:

- **A and B are exact.** B11 = −B22 with B66 = 0 is the coupling expected for an unsymmetric cross-ply: stretching the plate in x or y also makes it bend.
- **D is about 0.39% below the exact value.** This comes from splitting each ply into 8 layers. It is the D the model will actually use; running with `--n-sub 32` reduces the gap to about 0.02%.
- **v23 and transverse shear don't affect these numbers.** So the assumed v23 = 0.34 can't be checked this way.

No project files were changed. The full-precision matrix is saved as a CSV at `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-10-23-26--Tara-\b3f2b058-0ddf-4c33-9eea-6659d2adbd92\scratchpad\abd_layup_90_0_opensees.csv`.

User

Now use the agent TheoreticalABDGenerator to calculate the ABD matrix manually

Claude

The TheoreticalABDGenerator agent is now computing the ABD matrix from classical lamination theory for the [90/0] layup. When it finishes, I'll post the matrices here and compare them with the OpenSeesPy ABD.

Claude

Here is the hand-calculated ABD matrix for the [90/0] layup, from classical lamination theory. A and B match the OpenSeesPy result exactly, and D differs by 0.39%.

Ply stiffness (MPa). With v21 = v12·E2/E1 = 0.0242857:

- Q11 = 141165.62, Q22 = 10083.26, Q12 = 3428.31, Q66 = 6000.
- The 0° ply uses these values as they are. For the 90° ply, Q11 and Q22 swap places.

Through-thickness positions: the 90° ply runs from z = −1 to 0 mm and the 0° ply from z = 0 to 1 mm.

The ordering is the same as before: N = A ε + B κ and M = B ε + D κ.

A [N/mm]

	ε_xx	ε_yy	γ_xy
N_xx	151248.88	6856.62	0
N_yy	6856.62	151248.88	0
N_xy	0	0	12000

B [N]

	κ_xx	κ_yy	κ_xy
N_xx	65541.18	0	0
N_yy	0	-65541.18	0
N_xy	0	0	0

D [N·mm]

	κ_xx	κ_yy	κ_xy
M_xx	50416.29	2285.54	0
M_yy	2285.54	50416.29	0
M_xy	0	0	4000

Hand checks for this layup:

- A11 = Q11 + Q22.
- B11 = −B22 = ½(Q11 − Q22).

- D = A/3.

Comparison with the OpenSeesPy ABD:

Term	Hand calculation	OpenSeesPy	Difference
A11	151248.88	151248.88	~1e-16
A12	6856.62	6856.62	~1e-16
A66	12000	12000	0
B11 / B22	±65541.18	±65541.18	~1e-16
D11	50416.29	50219.36	−0.391%
D12	2285.54	2276.61	−0.391%
D66	4000	3984.375	−0.391%

Every D term is low by the same factor, exactly 1/256 = 1/16². That is the expected error from OpenSeesPy splitting the laminate into 16 layers (8 per ply) and evaluating each layer at its middle. So the OpenSeesPy layup is correct, and running with `--n-sub 32` would bring D within about 0.02%.

The full-precision matrix is saved as a CSV at `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-10-23-26--Tara-\b3f2b058-0ddf-4c33-9eea-6659d2adbd92\scratchpad\abd_clt.csv`. No project files were changed.

User

Now run the same workflow (generate OpenSeesPy code for plies using PlyPropertyGenerator, generate OpenSeesPy code for layup using LayupFromPlyGenerator, calculate and report ABD matrix based on OpenSeesPy code using ABDFromOpenSeesPyLayup, calculate and report ABD matrix from classical laminate theory using TheoreticalABDGenerator), but for the following inputs:

Material properties: E1 = 139.5 GPA, E2 = 6.101 GPa, nu12 = 0.2750, G12 = 2.574 GPa, nu21 = 0.0120, and ply thickness is 54 microns

The 4 following layups (where plies are listed from top to bottom):

1. [0/90/90/0]

2. [90/0/0/90]

3. [45/0/0/45]

4. [-45/90/90/-45]

Claude

Two agents are now running in parallel:

- **PlyPropertyGenerator** is building the 0°, 90°, 45° and −45° ply materials for the new CFRP in a new file, `ply_materials_139GPa.py`. The earlier 0°/90° files won't be touched.
- **TheoreticalABDGenerator** is computing the classical-lamination-theory ABD for all four layups. It doesn't depend on the OpenSeesPy code, so it can start now.

Decisions I made:

- **v21:** I'm using v21 = v12·E2/E1 = 0.012027 rather than your rounded 0.0120, so the material constants stay consistent with each other.
- **v23:** it isn't given, so it's assumed equal to v12. It doesn't affect the ABD matrix.
- **Ply order:** all four layups are symmetric, so listing them top-to-bottom gives the same order as bottom-to-top.

Once the ply materials are done, I'll have LayupFromPlyGenerator build the four layups. Then ABDFromOpenSeesPyLayup will measure each one's ABD, and I'll compare those with the hand-calculated values.

Claude

The hand-calculated (classical lamination theory) ABD matrices for all four layups are done. PlyPropertyGenerator is still building the ply materials. The OpenSeesPy layups and their ABD can't start until it finishes. Once they're done, I'll report both sets of matrices side by side.

Claude

The ply materials are done, in `ply_materials_139GPa.py`. All four angles (0°, 90°, 45°, −45°) passed the single-element stiffness check to within rounding error. LayupFromPlyGenerator is now writing the OpenSeesPy code for the four layups.

Ply materials:

- **Tags:** 0° is 11/12/13, 90° is 14/15/16, 45° is 17/18/19 and −45° is 20/21/22. They start at 11 so they don't clash with the earlier 0°/90° file.
- **Element orientation (`-local`):** (1,0,0) for 0°, (0,1,0) for 90°, (0.707, 0.707, 0) for 45° and (0.707, −0.707, 0) for −45°.
- **Assumptions:**
  - v21 = 0.0120271, from v12·E2/E1 so the constants stay consistent.
  - v23 = v12 = 0.275, which gives G23 = 2392.55 MPa.
  - Density is 0.

Hand-calculated (classical lamination theory) results, already done:

- **B = 0 for all four layups,** as expected since they're all symmetric.
- **L1 and L2:** they have the same A, but D differs because the outer plies dominate bending:
  - L1 (0° outside): D11 = 103.5 N·mm.
  - L2 (90° outside): D11 = 19.2 N·mm.
- **L3 and L4 have shear–extension and bend–twist coupling** (A16, A26, D16 and D26 aren't zero). Each one has an off-axis angle with no opposite-sign partner, so nothing cancels.

Next, ABDFromOpenSeesPyLayup will measure the ABD from each layup's code, and I'll compare all of them with the hand-calculated values.

Claude

The OpenSeesPy code for all four layups is written, and each script passes its own check against the hand-calculated ABD. ABDFromOpenSeesPyLayup is now measuring the ABD of each layup from that code. When it finishes, I'll post the full comparison.

New scripts:

- `layup_L1_0_90_90_0.py`
- `layup_L2_90_0_0_90.py`
- `layup_L3_45_0_0_45.py`
- `layup_L4_m45_90_90_m45.py`

Plies at the same angle share one element. Each mesh face therefore gets 2 stacked elements (one per distinct angle) instead of 4. The agent checked that this gives the same stiffness as using one element per ply, to rounding error, and it halves the element count.

Accuracy against the hand-calculated values:

- **A and B** match to rounding error, including the A16/A26 coupling terms in L3 and L4.
- **D** is within 0.06% for L1 and L2, and within 0.17% for L3 and L4.

Claude

All four layups went through the full workflow, and the OpenSeesPy ABD matrices agree with the hand-calculated ones (classical lamination theory). I compared the two sets of saved matrices myself, not just the agents' reports.

Inputs:

- **Moduli:** converted from GPa to MPa: E1 = 139500, E2 = 6101, G12 = 2574.
- **Plies:** each ply is t = 0.054 mm, so the total thickness h is 0.216 mm.
- **v21:** I used 0.0120271 (= v12·E2/E1) rather than your rounded 0.0120, so the constants stay consistent with each other.
- **v23:** assumed equal to v12 = 0.275. It doesn't affect the ABD.

The relations are N = A ε + B κ and M = B ε + D κ. Units are A [N/mm], B [N], D [N·mm]. **B = 0 for all four layups in both methods**, as expected since they're all symmetric.

L1 [0/90/90/0]

	11	22	12	66	16 = 26
A (both methods)	15777.1	15777.1	363.602	555.984	0
D, hand calculation	103.492	19.1908	1.41368	2.16167	0
D, OpenSeesPy	103.432	19.1309	1.4123	2.15955	0

L2 [90/0/0/90]

	11	22	12	66	16 = 26
A (both methods)	15777.1	15777.1	363.602	555.984	0
D, hand calculation	19.1908	103.492	1.41368	2.16167	0
D, OpenSeesPy	19.1309	103.432	1.4123	2.15955	0

L3 [45/0/0/45]

	11	22	12	66	16 = 26
A (both methods)	19429.2	4974.26	3938.98	4131.36	3613.73
D, hand calculation	44.0395	29.9894	25.7406	26.4886	24.5878
D, OpenSeesPy	43.9658	29.9705	25.7256	26.4729	24.5741

L4 [−45/90/90/−45]

	11	22	12	66	16 = 26
A (both methods)	4974.26	19429.2	3938.98	4131.36	−3613.73
D, hand calculation	29.9894	44.0395	25.7406	26.4886	−24.5878
D, OpenSeesPy	29.9705	43.9658	25.7256	26.4729	−24.5741

How well they agree

Layup	A (max relative difference)	B (largest value, should be 0)	D (max relative difference)
L1	2.3e-16	1.7e-12 N	0.058%
L2	3.5e-16	1.7e-12 N	0.058%
L3	3.7e-16	7e-13 N	0.17%
L4	1.6e-16	8e-13 N	0.17%

What the matrices show

- **D is slightly low in OpenSeesPy.** It splits each ply into 8 thin layers and evaluates each at its middle, which loses a little bending stiffness. You can tighten it by running a layup script with `--n-sub` set higher.
- **L1 and L2 have the same in-plane stiffness (A)** because both contain two 0° and two 90° plies. Their bending stiffness differs because the outer plies dominate bending: D11 is 103 N·mm with 0° plies outside and 19 N·mm with 90° plies outside.

- **L3 and L4 have coupling terms.** Stretching causes shear (A16, A26) and bending causes twisting (D16, D26). Each layup has a ±45° ply with no opposite-sign partner, so nothing cancels.
- **L4 is L3 rotated 90°.** The 11 and 22 terms trade places and the 16/26 terms flip sign.

Files

New in the project folder:

- `ply_materials_139GPa.py`, with material tags 11–22.
- `layup_L1_0_90_90_0.py`
- `layup_L2_90_0_0_90.py`
- `layup_L3_45_0_0_45.py`
- `layup_L4_m45_90_90_m45.py`

In each layup, plies at the same angle share one element, so each mesh face gets 2 stacked elements instead of 4. The elements must use `-noeas`, so refine the mesh until results stop changing. No existing files were changed.

The full-precision matrices are saved as CSVs in the scratchpad folder `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-10-23-26--Tara-\b3f2b058-0ddf-4c33-9eea-6659d2adbd92\scratchpad\`:

- Hand-calculated: `abd_clt_L1.csv` to `abd_clt_L4.csv`
- OpenSeesPy: `abd_opensees_L1.csv` to `abd_opensees_L4.csv`

User

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